cognizant

Digital Nurture 5.0

Deep Skilling Handbook - DotNet FSE Angular

Program Highlights

- The Deep Skilling Learning program runs for a period of 7 weeks. Go through the recommended self- learning resources and practice the exercises to excel in the recommended DotNet FSE modules.

- Deep Skilling covers the below mentioned concepts.

Recommended Program Sequence

The learning journey contains the modules mentioned above, followed by a Deep Skilling Final Assessment. It is recommended to follow a sequence of skills learning, the details of which are given below.

Reason for the sequence recommendation:

Any engineering competency landscape can be classified into four major constructs

1. **Engineering Concepts**

   The core concepts for any engineering landscape. This is the foundation on which all programming languages and frameworks are built.

   Relevant skills:

   - Design Patterns and Principles
   - Data Structures and Algorithms

2. **Programming Languages**

   The languages are used to implement the necessary logic for the requirements.

   Relevant skills:

   - C# - Covered in upskilling
   - SQL - Basic (covered in upskilling) and Advanced (covered in deep skilling)
   - NUnit and Moq

3. **Products and Frameworks**

   There are numerous frameworks in the programming family to build applications. These enable the programmers to develop different types of applications by providing a gamut of features provided over a single integrated environment.

   Relevant skills:

   - Entity Framework Core 8.0
   - ASP.NET Core 8.0 Web API
   - Microservices Architecture using ASP.NET Core Web API – This is an architecture pattern.
   - Angular

4. Platforms

These are the enablers as a base to host the application developed using the Engineering concepts, programming languages, product and frameworks to make it accessible by all users, reliable and scalable.

Relevant skills:

- GIT
- CI/CD
- Containerization using Docker

Apart from all the mentioned skills, the very widely emerging concentration that any industry is looking into is the 'Generative Artificial Intelligence' or the 'GenAI'.

Let us get an introduction to it. You can go through it in the module 'Gen AI Fundamentals'.

Program Completion Criteria

To successfully complete this learning program, candidates must meet the following criteria:

Hands-on Exercises:

- Candidates must complete the hands-on exercises mapped against every skill. These exercises are designed to reinforce learning objectives and ensure a practical understanding of the material.

Final KBA Assessment:

- Candidates must pass the final **Knowledge-Based Assessment (KBA)**. This assessment will evaluate their comprehensive understanding and application of the concepts covered throughout the program.

Learning Approach

DN 5.0 Deep Skilling Program adopts a comprehensive and blended learning approach to ensure an engaging and effective educational experience.

The program comprises two essential learning components:

- **Self-paced learning** through open-source learning reference links.
- **Bi-weekly SME connect** sessions conducted by experts.

Self-Paced Learning using Open-source Reference Links

- Please refer to the learning reference links provided to learn and understand the recommended concepts.
- **We expect you to dedicate 10-12 hours of focused attention weekly** to your learning modules.

Introduction to Cognizant SkillSpring

Cognizant SkillSpring is a multimodal, AI-native, conversational learning platform designed to redefine learning in the AI era and help businesses cultivate AI-ready talent at scale. Moving beyond static courses, SkillSpring uses AI-powered agents to personalize learning, accelerate skill mastery, and embed learning directly into the flow of work.

Candidates are highly encouraged to leverage the SkillSpring platform and the specific courses highlighted in the modules below to enhance your deep skilling journey and become future-ready.

You'd be enrolled to the relevant courses for which you'd receive an email on the process to access them and start learning. This would be done within 1.5 weeks of the Deep skilling Learning Journey.

SME Connect Session

- We will have scheduled sessions with Subject Matter Experts (SMEs) that are designed to deepen your understanding of complex topics.
- **100% attendance is mandatory** for SME Connect sessions.
- Engage actively in doubt clarification sessions, asking questions and seeking clarification on challenging topics.

Disclaimer: Cognizant does not claim ownership or responsibility for the content or any issues with the links provided, as they are merely references available on the internet. Candidates are free to leverage additional sources beyond what has been provided to enhance their skill capabilities for Deep Skilling.

Effective Learning Strategies

Create a Study Schedule	Set Clear Goals	Stay Organized	Active Learning
• Dedicate specific times each week for learning. • Aim for 10-12 hours of study per week.	• Define what you want to achieve each week. • Break down tasks into manageable chunks.	• Keep track of your progress and deadlines. • Use tools like calendars, to-do lists, and reminders.	• Engage with the material through hands-on exercises. • Practice coding and solving problems regularly.

Duration Recommendation

Weekly Learning: 10-12 hours	Total Program Duration: 7 weeks
• Allocate 1-2 hours daily or schedule longer sessions on weekends. • Balance your learning with regular academic commitments.	• Follow the recommendations consistently. • Ensure to complete the mandatory exercises and review sessions.

Where to Practice?

Installed Software Tools
• The skills covered require 'Visual studio community edition' to be used. Download it from Visual Studio Community | Download Latest Free Version • Install Sql server management studio from Install SQL Server Management Studio | Microsoft Learn • Install Visual studio code from Download Visual Studio Code - Mac, Linux, Windows • Install necessary tools and libraries as per course requirements.

Exercise Instructions

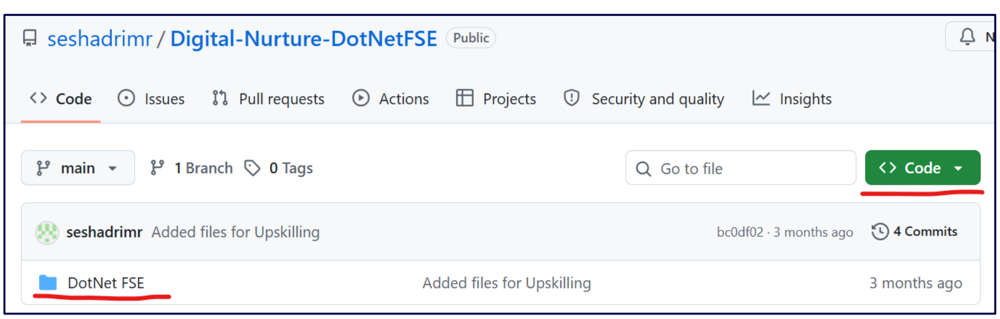
In this learning program, you will be required to complete weekly exercises designed to reinforce the concepts learned during the week. These exercises are hosted on a public GitHub repository and must be downloaded and solved on a weekly basis. Follow the instructions below to ensure a smooth and productive exercise workflow:

1. **Accessing Exercises:**

- Each week, exercises will be made available in our public GitHub repository.
- The repository URL is this.

2. **Downloading Exercises:**

- You'll see the options 'DotNet FSE' as shown in the screenshot below. Click the 'Code' button as shown in the screenshot and choose the option 'Download zip'. This will download all the required Hands-on content for your practice.



3. **Solving the Exercises:**

- You'll need to complete **Mandatory hands-on**, identified for every skill. The details of such hands-on is provided in the file 'DN - DotNet FSE Mandatory hands-on detail.xlsx' in the same GitHub repository. The remaining ones are additional ones, which is up to you to attempt them.
- Solve the problem statements provided in the downloaded files.
- Ensure that you understand the problem requirements and apply the concepts learned during the week.
- Take your time to think through the solutions and code them accurately.

4. **Self-Evaluation:**

- After completing the exercises, evaluate your solutions based on the problem criteria.
- Compare your approach with any hints or solutions provided if available.
- Reflect on any mistakes or areas where you can improve.

5. **Submitting Solutions:**

- Firstly, organize your solutions week-wise and keep them in a folder.
- Create a **public repository** in your personal GitHub account, upload your solution folder, and share the URL with the POC on demand.

6. **Additional Support:**

- If you encounter difficulties or have questions about the exercises, seek help from your peers.
- Utilize the resources and links provided in this handbook for further assistance.

Upskilling skills – A quick recap

Hi Champ – Welcome to the Deep skilling of Digital Nurture 5.0 program. Before you delve deep into the skills into this, with utmost sincerity, please do go through the upskilling handbook and refresh the basic programming skills to keep yourself abridged.

Always remember – The Object-Oriented Programming concepts are the core for any Programming language or frameworks.

Correlate the concepts you learn with real-life to understand and relate better for better understanding and life-long knowledge retention.

FSE construct - Engineering concepts

The core concepts for any engineering landscape. This is the foundation on which all programming languages and frameworks are built.

Relevant skills:

- Design Patterns and Principles
- Data Structures and Algorithms

Please plan to complete the skills in 2 days.

Module 1 - Design Patterns and Principles

Overview:

This module introduces learners to essential design principles and patterns that are crucial for creating robust and maintainable software. Learners will delve into the SOLID principles, which include SRP, OCP, LSP, ISP, and DIP. They will also explore common design patterns, such as Creational, Structural, and Behavioral patterns, which provide reusable solutions to common design challenges. The module combines theoretical insights with practical exercises, helping learners understand and apply these concepts in their projects, ultimately enhancing their ability to write clean, efficient, and scalable code.

Learning Objectives:

After completing this module, users will be able to:

- Grasp the importance of the Single Responsibility Principle (SRP).
- Implement the Open/Closed Principle (OCP) to create extendable software.
- Ensure software components adhere to the Liskov Substitution Principle (LSP).
- Design interfaces following the Interface Segregation Principle (ISP).
- Apply the Dependency Inversion Principle (DIP) for flexible dependency management.
- Recognize and apply Creational Patterns.
- Utilize Structural Patterns to build robust systems.
- Employ Behavioral Patterns for effective object interaction.
- Improve code reusability and reduce duplication through design patterns.
- Solve common software design problems using appropriate design patterns.
- Evaluate and choose the right design pattern for a given problem context.

Self-Learning (Open-source links):

Key Topics	Sub-topics	Learning Reference Links
SOLID Principles	What & Why, **The SOLID Principles of Object-Oriented Programming** - The Single Responsibility Principle, The Open-Closed Principle, The Liskov Substitution Principle, The Interface Segregation Principle, The Dependency Inversion Principle	https://www.baeldung.com/solid-principles
Commonly used Design Patterns	GoF, **Creational Patterns** - Singleton Pattern, Factory Method Pattern, Builder Pattern; **Structural Patterns** - Adapter Pattern, Decorator Pattern, Proxy Pattern; **Behavioral Patterns** - Observer Pattern, Strategy Pattern, Command Pattern; **Architectural Patterns** - Model-View-Controller (MVC), Dependency Injection	https://medium.com/@softwaretechsolution/design-pattern-81ef65829de2

Check Your Understanding:

Design Patterns Quiz ▶

Hands-On:

- Complete the skills' hands-on exercises to reinforce your learning. Refer to the **Exercise Instructions** section above to access the practice content.

Module 2 - Data Structures and Algorithms

Overview:

This module focuses on core data structures and algorithms, foundational for writing efficient and scalable code. Learners will explore key data structures such as arrays, linked lists, and understand their implementation and application scenarios. Additionally, they will delve into fundamental algorithms including searching (linear search, binary search) and sorting (bubble sort, quick sort, merge sort). The module emphasizes practical implementation through coding exercises, enabling learners to apply these concepts effectively in real-world programming challenges.

Learning Objectives:

After completing this module, learners will be able to:

- Choose appropriate data structures based on problem requirements.
- Apply searching algorithms to find elements in data structures.
- Utilize sorting algorithms to order data efficiently.
- Solve coding problems using appropriate data structures and algorithms.
- Analyze algorithmic complexities to optimize performance.
- Design efficient and scalable solutions for software development tasks using learned concepts.

Self-Learning (Open-source links):

Key Topics	Sub-topics	Learning Reference Links
Analysis of Algorithms	Introduction, Why DS& Algorithm, Types of DS, Notations, Time and Space Complexity, Start using frameworks for describing and analyzing algorithms, Begin using asymptotic notation to express running-time analysis, Asymptotic notations for run-time analysis of algorithms, Best Case, Average Case, Worst case analysis of an algorithm, Finding Time Complexity of few iterative and recursive algorithms	https://www.geeksforgeeks.org/design-and-analysis-of-algorithms/
Sorting	Bubble, Insertion, Heap Sort, Quick Sort, Merge Sort - Worst, Average and Best-Case analysis	https://www.geeksforgeeks.org/sorting-algorithms/
Arrays	Array Traversal - Array representation in Memory, Measuring Time complexity, Searching, Traversal in Arrays, when to use Arrays	https://www.geeksforgeeks.org/array-data-structure-guide/
Linked List	Single Linked List, Circular Single Linked List, Double Linked List, Circular Double Linked List - Search, Inert, Traverse, Delete operations, Time complexity	https://www.geeksforgeeks.org/linked-list-in-DotNet/
Searching	Linear Search, Binary Search	https://www.geeksforgeeks.org/searching-algorithms/#basics-of-searching-algorithms

Check Your Understanding:

Data Structures and Algorithms Quiz ▶

Hands-On:

- Complete the respective skill's hands-on exercises to reinforce your learning. Refer to the **Exercise Instructions** section above to access the practice content.

FSE construct - Programming Languages

The languages are used to implement the necessary logic for the requirements.

Relevant skills:

- C# - Covered in upskilling
- SQL – Basic (covered in upskilling) and Advanced (covered in deepskilling)

Please plan to complete this in 2 days.

Module 3 - Advanced SQL Using SQL Server

Overview:

This module covers a little advanced topics in SQL server for programming in it, apply logic to formulate Stored procedures and execute them, transactions to ensure data integrity.

Learning Objectives:

After completing this module, learners will be able to:

Self-Learning (Open-source links):

Key Topics	Sub-topics	Learning Reference Links
Advanced Concepts	Introduction to SQL Server, SQL Server Editions, Key Components and Services of SQL Server, SQL Server Instances, SQL Server Management Studio	https://www.geeksforgeeks.org/introduction-of-ms-sql-server/ https://learn.microsoft.com/en-us/ssms/download-sql-server-management-studio-ssms
	OVER(), PARTITION BY, ROW_NUMBER, RANK, DENSE_RANK, GROUPING SETS, CUBE, ROLLUP, WITH Statement, Common Table Expression (CTE), Recursive CTE, MERGE, PIVOT, UNPIVOT	https://www.sqlservertutorial.net/sql-server-window-functions/ https://www.sqlservertutorial.net/sql-server-basics/ https://www.geeksforgeeks.org/unpivot-in-sql-server/
Views and Indexes	Introduction to Views, Creating Views, Modifying Views, Dropping Views, Types of Views, Updating Views, Indexed Views, Security and Views, Introduction to Indexes, Types of Indexes, Creating Indexes, Modifying Indexes, Dropping Indexes, Covering Indexes, Index	https://www.sqlservertutorial.net/sql-server-views/ https://www.sqlservertutorial.net/sql-server-indexes/

	Fragmentation, Indexing and Query Optimization	https://www.mssqltips.com/sqlservertip/4331/sql-server-index-fragmentation-overview/ https://www.geeksforgeeks.org/best-practices-for-sql-query-optimizations/
Stored Procedures and User-Defined Functions	What is a Stored Procedure?, Benefits of Using Stored Procedures, Types of Stored Procedures - User-defined, Temporary, System, Extended User-Defined, Create a Stored Procedure, Modify a Stored Procedure, Delete a Stored Procedure, Execute a Stored Procedure, Grant Permissions on a Stored Procedure, Return Data from a Stored Procedure, Rename a Stored Procedure, Parameters, What are Functions in SQL Server?, Benefits of user-defined functions, Types of functions-Scalar functions, Table-valued functions, System functions, Built-in system functions, Create User-defined Functions, Modify User-defined Functions, Delete User-defined Functions, Execute User-defined Functions	https://www.sqlservertutorial.net/sql-server-stored-procedures/ https://learn.microsoft.com/en-us/sql/relational-databases/stored-procedures/grant-permissions-on-a-stored-procedure?view=sql-server-ver16 https://www.sqlservertutorial.net/sql-server-user-defined-functions/
Triggers and Cursors	What are Triggers?, When we use Triggers?, Types of SQL Server Triggers, DML Triggers - After Triggers,Instead Of Triggers, Logon Triggers, Create Trigger, Modify Triggers using SSMS, DELETE Triggers, Advantages of Triggers, Disadvantages of Triggers, What are Cursors?, Life Cycle of the Cursor, Uses of SQL Server Cursor, Types of Cursors in SQL Server, Limitations of SQL Server Cursor, Alternates to Cursor	https://www.sqlservertutorial.net/sql-server-triggers/ https://www.sqlservertutorial.net/sql-server-stored-procedures/sql-server-cursor/ https://www.scholarhat.com/tutorial/sqlserver/sql-server-cursor-alternatives
Exception Handling	What are Exceptions?, Types of SQL Server Exceptions - System Defined, User Defined, Retrieving detailed information on the error, TRY/CATCH, Using Nested TRY...CATCH Constructs, THROW and RAISERROR	https://www.mssqltips.com/sqlservertip/7997/sql-server-try-catch-raiserror-throw-error-handling/

Transactions	What are transactions?, ACID Properties, Transaction States, Transaction Management Statements, Savepoints, Implicit vs. Explicit Transactions, Nested Transactions, Transaction Isolation Levels, Locking and Blocking, Transaction Deadlocks	https://www.sqlservertutorial.net/sql-server-basics/sql-server-transaction/ https://www.geeksforgeeks.org/sql-server-transaction/ https://dotnettutorials.net/lesson/sql-server-savepoints-transaction/ https://www.sqlservercentral.com/articles/isolation-levels-in-sql-server https://www.sqlservertutorial.net/sql-server-administration/sql-server-deadlock/

Hands-On:

- Complete the respective skill's hands-on exercises to reinforce your learning. Refer to the **Exercise Instructions** section above to access the practice content.

FSE construct - Products and Frameworks

Products and Frameworks

There are numerous frameworks in the programming family to build applications. These enable the programmers to develop different types of applications by providing a gamut of features provided over a single integrated environment.

Please plan to complete this in 5 weeks' time.

Relevant skills:

- Nunit and Moq – 1 day
- Entity Framework Core 8.0 - 1 weeks
- ASP.NET Core 8.0 Web API – 1.5 weeks
- Microservices Architecture using ASP.NET Core Web API – This is an architecture pattern. - 0.5 week
- Angular – 2 weeks

Module 4 – NUnit and Moq

Overview:

This module focuses on the Unit testing concepts through Microsoft .Net's NUnit and Mocking framework.

Learning Objectives:

After completing this module, learners will be able to:
- Explain what automated testing is and its benefits.
- Demonstrate unit testing concepts using NUnit framework.
- Demonstrate the usage of mocking framework to mock the dependencies in unit testing completion.

Self-Learning (Open-source links):

Key Topics	Sub-topics	Learning Reference Links
Getting Started	What is Automated Testing, Benefits of Automated Testing, Types of Tests, Test Pyramid, Popular Testing Frameworks, Using NUnit in Visual Studio, What is Test-Driven Development	https://www.geeksforgeeks.org/automation-testing-software-testing/ https://www.geeksforgeeks.org/benefits-of-automation-testing/ https://martinfowler.com/articles/practical-test-pyramid.html https://www.geeksforgeeks.org/test-driven-development-tdd/
Fundamentals of Unit Testing	Characteristics of Good Unit Tests, What to Test and What Not to Test, Naming and Organizing Tests, Black-box Testing, Set Up and Tear Down, Parameterized Tests, Ignoring Tests, Writing Trustworthy Tests	https://www.tutorialspoint.com/unit-testing-tutorial-for-beginners-concepts-types-tools https://www.geeksforgeeks.org/software-engineering-black-box-testing/
Core Unit Testing Techniques	Testing Strings, Testing Arrays and Collections, Testing Return Type of Methods, Testing Void Methods, Testing Methods that Throw Exceptions, Testing Private Methods, Code Coverage	https://www.c-sharpcorner.com/article/introduction-to-nunit-testing-framework/ https://dotnetpattern.com/nunit-introduction

Breaking External Dependencies	Loosely-coupled and Testable Code, Refactoring Towards a Loosely coupled Design, Dependency Injection via Method Parameters, Dependency Injection via Properties, Dependency Injection via Constructor, Dependency Injection Frameworks, Mocking Frameworks, Creating Mock Objects Using Moq, State-based vs. Interaction Testing, Testing the Interaction Between Two Objects	https://dotnettutorials.net/lesson/dependency-injection-design-pattern-csharp/ https://dotnettutorials.net/lesson/dependency-injection-design-pattern/ https://dotnettutorials.net/lesson/unity-container-asp-net-mvc/ https://www.codemag.com/Article/2305041/Using-Moq-A-Simple-Guide-to-Mocking-for-.NET

Hands-On:

Complete the respective skill's hands-on exercises to reinforce your learning. Refer to the **Exercise Instructions** section above to access the practice content.

Module 5 - Entity Framework Core 8.0

Overview:

This module introduces the learner to the Object Relational Framework concept through Entity framework of Microsoft .NET.

Learning Objectives:

After completing this module, learners will be able to:

- Explain what ORM framework is, the features in Entity framework Core supporting the concepts of ORM.
- Demonstrate database model creation using EF core.
- Demonstrate the database operations via the model created; use LINQ queries to perform the operation.

Self-Learning (Open-source links):

Key Topics	Sub-topics	Learning Reference Links
Overview of EF Core 8 and .NET 8 Integration	What is ORM (Object-Relational Mapping)?, EF Core vs EF Framework: Key Differences, New Features in EF Core 8	https://dotnettutorials.net/lesson/entity-framework-core/
Setting up EF Core in a .NET 8 Project	Installing EF Core Packages via NuGet, Configuring DbContext, Connecting to SQL Server, Basic EF Core CLI Commands (Add Migration, Update Database)	https://dotnettutorials.net/lesson/install-entity-framework-core/ https://dotnettutorials.net/lesson/dbcontext-entity-framework-core/ https://dotnettutorials.net/lesson/database-connection-string-in-entity-framework-core/
Creating a Simple Database Model	Defining Entities and Relationships Primary Keys, Foreign Keys, and Navigation Properties, Code-First Approach Overview	https://dotnettutorials.net/lesson/relationships-in-entity-framework-core/ https://dotnettutorials.net/lesson/student-management-application-using-ef-core-code-first-approach/ https://www.c-sharpcorner.com/blogs/entity-framework-code-first-vs-database-first-approach
Performing Basic CRUD Operations	Inserting Records into the Database (AddAsync), Retrieving Data with Find, FirstOrDefault, and ToListAsync, Updating Records and Tracking Changes, Deleting Records (Remove vs DeleteRange)	https://dotnettutorials.net/lesson/crud-operations-in-entity-framework-core/
LINQ Queries in EF Core 8	Writing Queries Using Where, Select, and OrderBy, Projection into DTOs (Data Transfer Objects), Filtering and Aggregating Data, Asynchronous Queries with ToListAsync()	https://dotnettutorials.net/lesson/linq-to-entities-in-entity-framework-core/
EF Core Migrations and Database Updates	Adding, Removing, and Updating Migrations, Seeding Data during Migrations, Managing Database Schema Changes	https://www.entityframeworktutorial.net/efcore/entity-framework-core-migration.aspx

Handling Relationships and Data Loading	Eager, Lazy, and Explicit Loading Explained, Configuring One-to-One, One-to-Many, and Many-to-Many Relationships, Navigating Circular References	https://blog.jetbrains.com/dotnet/2023/09/21/eager-lazy-and-explicit-loading-with-entity-framework-core/ https://dotnettutorials.net/lesson/relationships-in-entity-framework-core/
Performance Optimizations and Best Practices	Query Caching and Tracking Behavior (AsNoTracking), Batch Processing and Bulk Operations, Handling Concurrency with RowVersion Columns, Using Compiled Queries for Performance Boost	https://dotnetevangelist.net/post/performance-optimization-tricks/

Hands-On:

- Complete the respective skill's hands-on exercises to reinforce your learning. Refer to the **Exercise Instructions** section above to access the practice content.

Module 6 - ASP.NET Core 8.0 Web API

Overview:

This module introduces the learner to the concepts of ASP. Net Core, RESTful services, its implementation in Microsoft .NET through Web Api.

Learning Objectives:

After completing this module, learners will be able to:

- Explain the concept of RESTful web service, Web API & Microservice
- List the types of Action Verbs
- Demonstrate creation of a simple Web API - With Read, Write actions
- Demonstrate Swagger installation to Web API and Web API listing on browser
- Demonstrate JWT-based authentication and authorization

Self-Learning (Open-source links):

Skill Spring Recommended Course:

- Course Name: .NET Backend Engineer

Key Topics	Sub-topics	Learning Reference Links
Introduction to Web APIs and ASP.NET Core	Overview of Web APIs, REST vs. SOAP, Setting up .NET 8 Development Environment, ASP.NET Core Web API project structure	https://dotnettutorials.net/lesson/introduction-to-asp-net-core-web-api/ https://dotnettutorials.net/lesson/environment-setup-asp-net-core-web-api/ https://dotnettutorials.net/lesson/asp-net-core-web-api-project-in-visual-studio-2019/ https://dotnettutorials.net/lesson/asp-net-core-web-api-files-and-folders/
Building RESTful APIs with ASP.NET Core	Creating controllers, actions, and routes, CRUD operations with Entity Framework Core, JSON formatting and serialization	https://dotnettutorials.net/lesson/controllers-in-asp-net-core-web-api/ https://dotnettutorials.net/lesson/models-in-asp-net-core-web-api/ https://dotnettutorials.net/lesson/routing-in-asp-net-core-web-api/ https://dotnettutorials.net/lesson/get-method-in-web-api/ https://dotnettutorials.net/lesson/post-method-in-web-api/ https://dotnettutorials.net/lesson/put-method-in-web-api/ https://dotnettutorials.net/lesson/delete-method-in-web-api/
Advanced API Features	Attribute routing and query parameters, Middleware and custom filters, Implementing JWT-based authentication and authorization	https://dotnettutorials.net/lesson/variables-and-query-strings-in-routing/ https://dotnettutorials.net/lesson/filters-in-asp-net-core-web-api/ https://dotnettutorials.net/lesson/action-filters-in-asp-net-core-web-api/

		https://dotnettutorials.net/lesson/authorization-filters-in-asp-net-core-web-api/ https://dotnettutorials.net/lesson/jwt-authentication-in-asp-net-core-web-api/ https://dotnettutorials.net/lesson/role-based-jwt-authentication-in-asp-net-core-web-api/
Consuming and Creating SOAP Services	Introduction to SOAP APIs and WCF, Consuming SOAP services in ASP.NET Core, Creating SOAP services using WCF	https://www.tutorialspoint.com/soap/what_is_soap.htm https://www.tutorialspoint.com/wcf/wcf_overview.htm https://www.geeksforgeeks.org/difference-between-wcf-and-web-service/
API Security and Exception Handling	Global exception handling and error responses, Logging with Serilog, Securing APIs with API keys and Oauth	https://dotnettutorials.net/lesson/exception-filters-in-asp-net-core-web-api/ https://dotnettutorials.net/lesson/cors-in-asp-net-core-web-api/ https://dotnettutorials.net/lesson/logging-in-asp-net-core-web-api/ https://dotnettutorials.net/lesson/custom-logging-provider-in-asp-net-core-web-api/ https://dotnettutorials.net/lesson/logging-using-serilog-in-asp-net-core-web-api/ https://www.c-sharpcorner.com/article/using-api-key-authentication-to-secure-asp-net-core-web-api/
API Documentation and Testing	Integrating Swagger/OpenAPI for API documentation, Using Postman for manual API testing, Automating API testing with REST Client	https://dotnettutorials.net/lesson/swagger-api-in-asp-net-core-web-api/ https://www.softwaretestingmaterial.com/postman-tutorial/

Hands-On:

- Complete the respective skill's hands-on exercises to reinforce your learning. Refer to the **Exercise Instructions** section above to access the practice content.

Module 7 - Microservices Architecture using ASP.NET Core Web API

Overview:

This module focuses on the importance of loose coupling and logically independent services following design principles. The architectural pattern of Microservices using ASP. Net core Web Api.

Learning Objectives:

After completing this module, learners will be able to:

- Explain the architecture pattern of Microservices.
- Explain the advantages and challenges of Microservices.
- Compare the Microservices and Monolith architecture patterns.
- Demonstrate the service discovery pattern in ASP .NET Core Web Api.
- Explain the importance of monitoring and logging in microservices.
- Explain the deployment strategies for microservices.
- Explain an overview of automating deployment with Docker and Kubernetes.

Self-Learning (Open-source links):

Key Topics	Sub-topics	Learning Reference Links
Introduction to Microservices Architecture	Overview of microservices architecture, Advantages and challenges of microservices, Comparison with monolithic architecture, Setting up an ASP.NET Core Web API project	https://dotnettutorials.net/lesson/microservices-using-asp-net-core/
Microservice Communication and Data Management	Inter-service communication methods (HTTP, messaging, gRPC), Service discovery and registration Implementing communication patterns in ASP.NET Core Web API, Database per service vs. shared database patterns, Implementing data access with Entity Framework Core in microservices, Data consistency and transactions in microservices	https://www.geeksforgeeks.org/inter-service-communication-in-microservices/
Security, Monitoring, and Deployment	Authentication and authorization in microservices, Implementing JWT (JSON Web Tokens) authentication, Securing ASP.NET Core Web API endpoints, Importance of monitoring and logging in microservices,	https://dotnettutorials.net/lesson/jwt-authentication-in-asp-net-core-web-api/ https://dotnettutorials.net/lesson/role-based-jwt-authentication-in-asp-net-core-web-api/ https://dotnettutorials.net/lesson/logging-in-asp-net-core-web-api/ https://medium.com/@jeslurrahman/implementing-health-checks-in-net-8-c3ba10af83c3

	Logging strategies in ASP.NET Core Web API, implementing health checks and metrics endpoints, Deployment strategies for microservices (rolling updates, blue-green deployments), Setting up CI/CD pipelines with Azure DevOps or GitHub Actions, Automating deployment with Docker and Kubernetes	https://medium.com/@hitendra.patel2986/deployment-strategies-for-microservices-a-complete-guide-8f403e6d8b3e https://medium.com/@sarshar.samoo/getting-started-with-ci-cd-pipelines-for-asp-net-core-api-using-azure-devops-1d637afd4e1f https://medium.com/@ucheblessed/ci-cd-with-docker-and-kubernetes-deploying-containerized-applications-7556f0727517

Hands-On:

- Complete the respective skill's hands-on exercises to reinforce your learning. Refer to the **Exercise Instructions** section above to access the practice content.

Module 8 – Angular (v20.0)

Overview:

This module provides a comprehensive introduction to Angular (v20.0), covering core concepts required to build modern, dynamic web applications. Learners will explore Angular architecture, components, directives, forms, services, routing, HTTP communication, state management, reactive programming with RxJS, and testing practices. The module emphasizes hands-on development using Angular CLI and industry best practices for scalable application development.

Learning Objectives:

After completing this module, learners will be able to:

- Explain Angular architecture, project structure, and environment setup using Angular CLI
- Develop and manage Angular components, data binding, and component lifecycle
- Use directives and pipes to control UI behavior and data transformation
- Build and validate forms using template-driven and reactive approaches
- Implement dependency injection and create reusable services
- Configure routing, navigation, and guards for multi-page applications
- Integrate APIs using HttpClient and handle asynchronous data with Observables
- Apply state management techniques using services and NgRx

- Use RxJS operators for reactive programming in Angular
- Perform unit and basic end-to-end testing for Angular applications

Self-Learning (Open-source links):

Key Topics	Sub-topics	Learning Reference Links
Introduction to Angular & Environment Setup	Overview of Angular framework Installing Node.js and Angular CLI (npm install -g @angular/cli) Creating a new Angular project (ng new) Angular project structure and key files Running (ng serve) and Building (ng build) applications	https://www.geeksforgeeks.org/angular-js/angular-tutorial/ https://www.freecodecamp.org/news/angular-for-beginners-course/ https://www.geeksforgeeks.org/how-to-learn-angular-in-2024/
Angular Components	Creating Components (ng generate component) Component Interaction – Data Binding: • Property Binding [property]="value" • Event Binding (event)="handler()" • Two-way Binding [(ngModel)]="property" Component Lifecycle Hooks (ngOnInit, ngOnChanges, ngOnDestroy etc.) Parent-Child Communication using @Input and @Output decorators	https://www.geeksforgeeks.org/angular-js/angular-tutorial/ https://www.geeksforgeeks.org/angular-js/angular-cheat-sheet-a-basic-guide-to-angular/ https://www.geeksforgeeks.org/angular-js/component-lifecycle-in-angular/ https://www.tutorialspoint.com/angular/angular-lifecycle-hooks.htm https://www.scholarhat.com/tutorial/angular/angular-component-lifecycle-hooks-explained
Directives and Pipes	Built-in Structural Directives: *ngIf, *ngFor, *ngSwitch Attribute Directives: ngClass, ngStyle, ngModel Creating Custom	https://www.geeksforgeeks.org/angular-js/angular-tutorial/ https://www.geeksforgeeks.org/angular-js/angular-cheat-sheet-a-basic-guide-to-angular/ https://www.w3schools.com/angular/angular_lifecycle.asp

	Directives Built-in Pipes: date, uppercase, currency, number Creating Custom Pipes	
Angular Forms	Template-driven Forms: • Basics, ngModel binding and Form Validation • Handling Form Submission Reactive Forms: • Setting up with FormBuilder • FormControl, FormGroup and FormArray • Built-in Validators and Custom Validators	https://www.geeksforgeeks.org/angular-js/angular-tutorial/ https://www.geeksforgeeks.org/angular-js/angular-cheat-sheet-a-basic-guide-to-angular/ https://www.tutorialspoint.com/angular/index.htm
Dependency Injection and Services	Introduction to Dependency Injection (DI) Creating and Using Angular Services (@Injectable) Hierarchical Dependency Injection (root, module, component level)	https://www.geeksforgeeks.org/angular-js/angular-cheat-sheet-a-basic-guide-to-angular/ https://www.tutorialspoint.com/angular/angular-dependency-injection.htm https://www.geeksforgeeks.org/how-to-learn-angular-in-2024/
Angular Routing and Navigation	Setting Up Routing – Configuring Routes in app-routing.module.ts Route Parameters and Query Parameters Nested Routes Lazy Loading Modules Router Guards: CanActivate, CanDeactivate, Resolve Router Events and Navigation Lifecycle Passing Data Between Routes	https://www.geeksforgeeks.org/angular-js/angular-tutorial/ https://www.geeksforgeeks.org/angular-js/implementing-lazy-loading-in-angular/ https://v18.angular.dev/guide/ngmodules/lazy-loading/

HTTP Client and APIs	Setting Up HttpClientModule Making HTTP Requests: GET, POST, PUT, DELETE Handling API Responses using Observables and Promises Error Handling and Retry Strategies HTTP Interceptors for modifying Requests and Responses • Adding auth headers globally, error handling, loading spinners, logging	https://www.geeksforgeeks.org/angular-js/angular-tutorial/ https://www.geeksforgeeks.org/angular-js/angular-cheat-sheet-a-basic-guide-to-angular/ https://www.geeksforgeeks.org/how-to-learn-angular-in-2024/ https://www.geeksforgeeks.org/angular-js/http-interceptors-in-angular/ https://www.geeksforgeeks.org/http-interceptor-use-cases-in-angular/
State Management in Angular	Introduction to State Management problem Using Services for State Management NgRx for Advanced State Management: • Store, Actions, Reducers, Effects, Selectors	https://www.geeksforgeeks.org/how-to-learn-angular-in-2024/ https://this-is-angular.github.io/ngrx-essentials-course/docs/chapter-12/
Reactive Programming with RxJS	RxJS Observables – what they are and why Angular uses them Common RxJS Operators: map, filter, switchMap, mergeMap, catchError Reactive Patterns in Angular	https://www.geeksforgeeks.org/angular-js/angular-tutorial/ https://www.geeksforgeeks.org/angular-js/angular-cheat-sheet-a-basic-guide-to-angular/
Testing Angular Applications	Unit Testing with Jasmine and Karma Component Testing and Service Testing End-to-End (E2E) Testing overview	https://www.geeksforgeeks.org/angular-js/angular-tutorial/ https://www.geeksforgeeks.org/angular-js/angular-cheat-sheet-a-basic-guide-to-angular/

Hands-On:

- Complete the respective skill's hands-on exercises to reinforce your learning. Refer to the **Exercise Instructions** section above to access the practice content.

Check your understanding:

- https://www.javaguides.net/2023/09/angular-quiz.html
- https://www.w3schools.com/angular/angular_quiz.asp
- https://www.geeksforgeeks.org/angular-js/angular-exercises-practice-questions-and-solutions/

Module 9 – Application Debugging

Overview:

This module focuses on debugging Angular applications using widely adopted tools such as Chrome DevTools and Visual Studio Code. Learners will gain practical experience in identifying, analyzing, and resolving frontend issues by inspecting the DOM, debugging TypeScript code, and examining application state. The module emphasizes efficient debugging practices to improve code quality and application performance.

Learning Objectives:

After completing this module, learners will be able to:

- Debug Angular applications using Chrome DevTools, including DOM inspection and breakpoints
- Analyze JavaScript and TypeScript code using the Sources panel and source maps
- Inspect elements and understand application structure through the DOM tree
- Configure and use Visual Studio Code debugger with Angular applications
- Set breakpoints, watches, and inspect variables in TypeScript files
- Debug Angular state management by inspecting services and NgRx store
- Apply effective debugging techniques to identify and fix frontend issues
- Improve application reliability and performance through systematic debugging

Self-Learning (Open-source links):

Key Topics	Sub-topics	Learning Reference Links
Debugging Angular with Chrome DevTools	Setting Up the Environment (ng serve) Opening Chrome DevTools (F12) Inspecting Elements (DOM tree) Debugging JavaScript – Sources panel, breakpoints, call stack Debugging Angular-specific code (TypeScript source maps)	https://www.geeksforgeeks.org/javascript/browser-developer-tools/ https://www.geeksforgeeks.org/javascript/debugging-in-javascript/ https://javascript.info/debugging-chrome https://buddy.works/tutorials/debugging-javascript-efficiently-with-chrome-devtools
Debugging Angular in Visual Studio Code	Installing Angular language extension for VS Code Attaching VS Code Debugger (launch.json configuration) Setting Breakpoints and Watches in TypeScript files Debugging State Management (inspecting services and NgRx store)	https://javascript.plainenglish.io/debug-angular-like-a-pro-stop-using-console-logs-and-start-using-vs-code-debugger-f1c6c001eac1 https://medium.com/@iReal_Nirmal/debug-angular-like-pro-with-visual-studio-code-54522ca923f1 https://info-12369.medium.com/how-to-debug-angular-applications-in-visual-studio-code-39b0acc417a8

Check your understanding:

- https://www.geeksforgeeks.org/quizzes/error-handling-and-debugging/
- https://www.toolsrail.com/quiz/web-browsers-developer-tools-quiz.php
- https://wayground.com/admin/quiz/66f797ec1ff0b846272b8a4e/mastering-chrome-dev-tools

FSE construct – Platforms & GenAI

Platforms

These are the enablers as a base to host the application developed using the Engineering concepts, programming languages, product and frameworks to make it accessible by all users, reliable and scalable.

Relevant skills:

- GIT
- CI/CD
- Containerization using Docker
- GenAI fundamentals

Please plan to complete this in 1 week.

Module 10 – Version control - GIT

Overview:

This module focuses on the most widely used code repository tool, GIT.

Learning Objectives:

After completing this module, learners will be able to:

- Explain the version control concepts.
- Demonstrate the branch clone, code push, forking operations
- Explain the concept of branch workflows

Self-Learning (Open-source links):

Key Topics	Sub-topics	Learning Reference Links
Introduction to Version Control	**Version Control Concepts**-Definition and Purpose, Benefits of Version Control, Types of Version Control Systems	https://www.geeksforgeeks.org/git-tutorial/ https://www.geeksforgeeks.org/version-control-systems/
Understanding Git	**What is Git**-Introduction to Git, Git as a Distributed Version Control System (DVCS); **Git Components**-Working Directory, Staging Area, Repository	https://www.geeksforgeeks.org/version-control-systems/ https://www.geeksforgeeks.org/git-introduction/ https://www.geeksforgeeks.org/what-is-a-git-repository/
Setting Up Git	**Installing Git**-Download and Installation Steps, Configuring Basic Settings (username, email); **Creating a Git Repository**-Initializing a New Repository, Cloning an Existing Repository;	https://www.geeksforgeeks.org/git-introduction/

Basic Git Commands	**git init and git clone** - Initializing a New Repository, Cloning an Existing Repository; **git add**-Staging Changes, Using Wildcards; **git commit**-Committing Changes, Adding Commit Messages; **git status**-Checking the Status of Your Repository; **git log**-Viewing Commit History, Options for Customizing Log Output;	https://www.geeksforgeeks.org/basic-git-commands-with-examples/ https://www.geeksforgeeks.org/essential-git-commands/
Branching and Merging	**Introduction to Branching**-Creating and Managing Branches, Switching Between Branches; **Merging Changes**-Merging Branches, Handling Merge Conflicts; **Branching Strategies**-Feature Branching, Release Branching, Git Flow Workflow;	https://www.geeksforgeeks.org/branching-strategies-in-git/ https://www.geeksforgeeks.org/git-merge/ https://www.geeksforgeeks.org/how-to-merge-two-branches-in-git/
Remote Repositories	**Adding Remote Repositories**-Linking Local and Remote Repositories, Multiple Remotes; **git pull and git push**-Pulling Changes from a Remote Repository, Pushing Changes to a Remote Repository; **Handling Remote Branches**-Tracking and Creating Remote Branches;	https://www.geeksforgeeks.org/what-is-a-git-repository/ https://www.geeksforgeeks.org/what-is-a-git-repository/#git-push-and-pull-commands
Collaborating with Git	**Forking and Pull Requests**-Forking a Repository, Creating and Managing Pull Requests; **Git Collaboration Workflows**-Centralized Workflow, Feature Branch Workflow, Forking Workflow, Gitflow Workflow	https://www.geeksforgeeks.org/git-fork/ https://www.geeksforgeeks.org/git-workflows-for-agile-development-teams/

Hands-On:

Complete the respective skill's hands-on exercises to reinforce your learning. Refer to the **Exercise Instructions** section above to access the practice content.

Module 11 – DevOps and CICD

Overview:

This module focuses on an awareness of the concept of Development and Operations done together in the current business development model.

Learning Objectives:

After completing this module, learners will be able to:

- Explain the concepts of DevOps
- Explain the details of the components of DevOps like Continuous Integration and Continuous Delivery
- List the popular tools used in DevOps

Self-Learning (Open-source links):

Key Topics	Sub-topics	Learning Reference Links
Introduction to DevOps	What is DevOps?, Goals and Benefits of DevOps, Key DevOps Practices	https://www.geeksforgeeks.org/devops-tutorial/ https://www.geeksforgeeks.org/introduction-to-devops/
Understanding CI/CD	What is Continuous Integration (CI)?, What is Continuous Deployment/Delivery (CD)?, Differences between CI and CD, Benefits of CI/CD	https://www.geeksforgeeks.org/introduction-to-devops/ https://www.geeksforgeeks.org/what-is-ci-cd/
CI/CD Tools and Platforms	Overview of Popular CI/CD Tools (e.g., Jenkins, GitHub Actions, GitLab CI/CD, CircleCI)	https://www.geeksforgeeks.org/introduction-to-devops/

Check your understanding:

DevOps quiz ▶

Module 12 – Containerization using Docker

Overview:

This module focuses on an awareness of the most widely used application containerization tool, Docker.

Learning Objectives:

After completing this module, learners will be able to:

- Explain the concept and features of Docker.
- List the basic Docker commands.
- Explain the concepts of Docker storage, networking, and orchestration.

Self-Learning (Open-source links):

Key Topics	Sub-topics	Learning Reference Links
Docker Commands	Basic Docker Commands - Run, ps, stop, rm, images, rmi, pull, Append a command, Exec	https://www.geeksforgeeks.org/docker-tutorial/
Docker Run	How to Use the docker run Command, Run a Container Under a Specific Name, Run a Container in the Background (Detached Mode), Run a Container Interactively, Run a Container and Publish Container Ports, Run a Docker Container and Remove it Once the Process is Complete	https://www.geeksforgeeks.org/docker-tutorial/
Docker Images	What is a Docker Image?, Image Layers, Container Layer, Parent Image, Base Image, Docker Manifest, Container Registries, Container Repositories, How to Create a Docker Image - Interactive Method, Dockerfile Method, The Docker Build Context	https://www.geeksforgeeks.org/docker-tutorial/

Docker Compose	What is Docker Compose?, Benefits of Docker Compose, Basic Commands in Docker Compose, Install Docker Compose, Create the Compose file,The YAML Configuration file	https://www.geeksforgeeks.org/docker-tutorial/
Docker Engine	What is Docker Engine- Docker CLI, REST API, Docker Deamon	https://www.geeksforgeeks.org/docker-tutorial/
Docker Storage	Storage Drivers, Data Volumes, Changing the Storage Driver for a Container, Creating a Volume, Listing all the Volumes	https://www.geeksforgeeks.org/docker-tutorial/
Docker Networking	Default Networks, Listing All Docker Networks, Inspecting a Docker network, Creating Your Own New Network	https://www.geeksforgeeks.org/docker-tutorial/
Container Orchestration	What is container orchestration?, Why do we need container orchestration?, What are the benefits of container orchestration?, What is Kubernetes container orchestration?, Container orchestration versus Docker	https://www.geeksforgeeks.org/docker-tutorial/

Check your understanding:

Docker quiz ▶

Module 13 – Agile Methodology

Overview:

This module provides a comprehensive understanding of Agile methodology, focusing on its core principles, frameworks, and practical implementation in software development. Learners will explore the Agile Manifesto, Scrum framework, estimation and planning techniques, and the creation of effective user stories. The module emphasizes collaboration, iterative development, and delivering value through Agile best practices.

Learning Objectives:

After completing this module, learners will be able to:

- Explain Agile principles, values, and the Agile Manifesto
- Differentiate between Agile and traditional Waterfall methodologies
- Describe Scrum roles, ceremonies, and artifacts in a project lifecycle
- Apply Agile estimation techniques such as story points and planning poker
- Interpret velocity and burndown charts for tracking progress
- Understand the sprint planning process and iterative delivery approach
- Write effective user stories using the standard format and INVEST principle
- Define and apply acceptance criteria using Given-When-Then format
- Collaborate effectively within Agile teams to deliver incremental value

Self-Learning (Open-source links):

Skill Spring Recommended Course:

- Course Name: ITPM - Introduction to Agile [101-Basics]

Key Topics	Sub-topics	Learning Reference Links
Introduction to Agile & Agile Manifesto	Overview of Agile principles and values Agile Manifesto – 4 values & 12 principles Agile vs Waterfall	https://www.geeksforgeeks.org/software-testing/what-is-agile-methodology/ https://www.atlassian.com/agile https://www.geeksforgeeks.org/software-engineering/agile-methodology-tutorial/
Scrum Framework – Roles, Ceremonies & Artifacts	Scrum roles: Scrum Master, Product Owner, Dev Team Scrum ceremonies: Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective Scrum artifacts: Product Backlog, Sprint Backlog, Increment Definition of Done	https://www.geeksforgeeks.org/software-engineering/what-is-scrum/ https://www.atlassian.com/agile/scrum https://www.atlassian.com/blog/project-management/beginners-guide-scrum-and-agile-project-management

Agile Estimation & Planning Techniques	Story Points concept Planning Poker technique Sprint planning process Velocity and burndown charts	https://www.tutorialspoint.com/estimation_techniques/estimation_techniques_planning_poker.htm https://www.mountaingoatsoftware.com/agile/planning-poker https://guide.freecodecamp.org/agile/planning-poker
Agile User Stories	What is a User Story? Format: As a [user], I want [goal], so that [benefit] INVEST principle Acceptance Criteria – writing effective criteria (Given-When-Then) Writing User Stories in practice	https://www.geeksforgeeks.org/software-engineering/user-stories-in-agile-software-development/ https://www.atlassian.com/agile/project-management/user-stories https://www.freecodecamp.org/news/how-to-write-user-stories-for-beginners https://www.geeksforgeeks.org/product-management/what-is-acceptance-criteria-how-to-write-it/

Module 14 – Gen AI Fundamentals

Overview:

This module introduces the fundamentals of Generative AI along with practical usage of GitHub Copilot as an AI-powered coding assistant. It covers key concepts such as prompt engineering techniques, ethical considerations, and real-world applications of GenAI in software development. Learners will gain hands-on experience in setting up Copilot, leveraging its core features, and understanding the responsibilities and risks associated with AI-assisted coding.

Learning Objectives:

After completing this module, learners will be able to:

- Explain the fundamentals, evolution, and applications of Generative AI
- Differentiate between generative AI and traditional AI approaches
- Apply prompt engineering techniques (zero-shot, few-shot, chain-of-thought) effectively
- Follow best practices and ethical guidelines for prompt design and AI usage
- Describe GitHub Copilot features, working principles, and supported environments
- Set up and configure GitHub Copilot in a development environment
- Use Copilot to generate code, documentation, and test cases efficiently
- Refactor and optimize code using AI-assisted suggestions
- Identify risks related to AI-generated code including security, licensing, and privacy concerns
- Apply responsible and secure practices while using AI coding tools

Self-Learning (Open-source links):

Skill Spring Recommended Course:

- Course Name: Prompt Engineering Foundation [101-BASIC]
- Course Name: GitHub Copilot Foundation Virtual Training Course

Key Topics	Sub-topics	Learning Reference Links
Introduction to Generative AI	What is Generative AI? How GenAI differs from traditional (discriminative) AI Overview of GenAI applications: text generation, code completion, image creation, chatbots History and evolution of Generative AI: • Early chatbots (1960s) → GANs (2014) → GPT-3 (2020) → ChatGPT (2022) → GitHub Copilot and beyond	https://www.geeksforgeeks.org/artificial-intelligence/what-is-generative-ai/ https://www.geeksforgeeks.org/what-is-generative-ai/ https://businessmanagementblog.com/history-of-generative-ai/
Prompt Engineering – Techniques, Best Practices & Ethics	What is Prompt Engineering and why it matters for developers Prompting Techniques: • Zero-shot prompting – direct task without examples • Few-shot prompting – giving examples to guide the model • Chain-of-thought prompting – asking model to reason step by step Best practices: clear instructions, provide context, specify output format, iterate Ethical considerations: avoiding bias in prompts, accuracy, privacy, responsible AI usage Hands-on: writing coding-task prompts (e.g., "Write a C# method that...")	https://www.geeksforgeeks.org/what-is-an-ai-prompt-engineering/ https://www.geeksforgeeks.org/blogs/prompt-engineering-best-practices/ https://www.promptingguide.ai/ https://www.codecademy.com/article/prompt-engineering-101-understanding-zero-shot-one-shot-and-few-shot https://www.tutorialspoint.com/prompt_engineering/prompt_engineering_ethical_considerations.htm
Introduction to GitHub Copilot	What is GitHub Copilot? Overview of GitHub Copilot features How Copilot works (AI pair programmer powered by OpenAI Codex) Supported IDEs and languages	https://www.geeksforgeeks.org/git/github-copilot/ https://www.freecodecamp.org/news/how-to-use-github-copilot-with-visual-studio-code/

Setup and Configuration	Installing GitHub Copilot extension in VS Code Connecting to a GitHub account Beginner-friendly first coding task with Copilot	https://www.freecodecamp.org/news/how-to-use-github-copilot-with-visual-studio-code/ https://www.geeksforgeeks.org/git/github-copilot/
Core Features and Capabilities	Code suggestions and completions (Tab to accept) Writing functions and boilerplate code from comments Generating comments and documentation automatically Refactoring and optimizing existing code Creating test cases with Copilot	https://www.geeksforgeeks.org/git/github-copilot/ https://www.freecodecamp.org/news/how-to-use-github-copilot-with-visual-studio-code/
Security and Ethical Considerations	Understanding AI-generated code risks (vulnerabilities, hallucinations) Licensing and attribution concerns (copyleft risk) Data privacy and usage policies (what code is sent to GitHub servers) Responsible use of Copilot – best practices	https://blog.gitguardian.com/github-copilot-security-and-privacy/ https://www.geeksforgeeks.org/git/github-copilot/ https://docs.github.com/en/copilot/responsible-use/code-review

Check your understanding:

- https://www.mygreatlearning.com/blog/generative-ai-quiz/
- https://aitoolsnote.com/quiz-prompt-engineering-mcq/
- https://technicalblog.in/languagewise-mcqs/prompt-engineering-mcqs/
- https://examsland.com/free-practice-test/github-copilot/

What's Next?

Congratulations on successfully completing the 7-week DN 5.0 Deep Skilling learning program!

As you have now finished this important phase of your learning, you will be taking a **Knowledge-Based Assessment (KBA)** to certify your skills. **This assessment will cover all the skills and topics you have learnt in this tenure**, ensuring you have a comprehensive understanding of the material.

We wish you the best of luck with your assessment and look forward to seeing you apply your newly acquired knowledge and skills. Good luck!